Q1: 24 Feb (Shift 1) - Numerical

If one of the diameters o of the circle $x^2+y^2-2x-6y+6=0$ is a chord of another circle ' C' whose center is at $(2,1)$, then its radius is

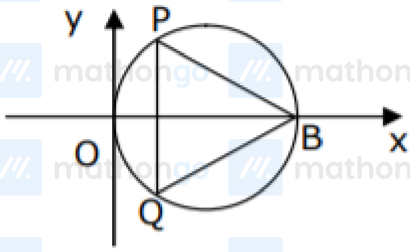
Q2: 24 Feb (Shift 2) - Numerical

If the area of the triangle formed by the positive x -axis, the normal and the tangent to the circle $(x-2)^2+(y-3)^2=25$ at the point (5,7) is A, then 24 A is equal to

Note: NTA has dropped this question in the final official answer key.

Q3: 26 Feb (Shift 1) - Single Correct

In the circle given below, let $OA=1$ unit, $OB=13$ unit and $PQ \perp OB$. Then, the area of the triangle PQB (in square units) is:



(1) $26\sqrt{3}$

(2) $24\sqrt{2}$

(3) $24\sqrt{3}$

(4) $26\sqrt{2}$

Q4: 26 Feb (Shift 2) - Single Correct

Let $A(1,4)$ and $B(1,-5)$ be two points. Let P be a point on the circle $(x-1)^2+(y-1)^2=1$ such that $(PA)^2+(PB)^2$ have maximum value, then the points, P, A and B lie on :

(1) a parabola

(2) a straight line

(3) a hyperbola

(4) an ellipse

Answer Key

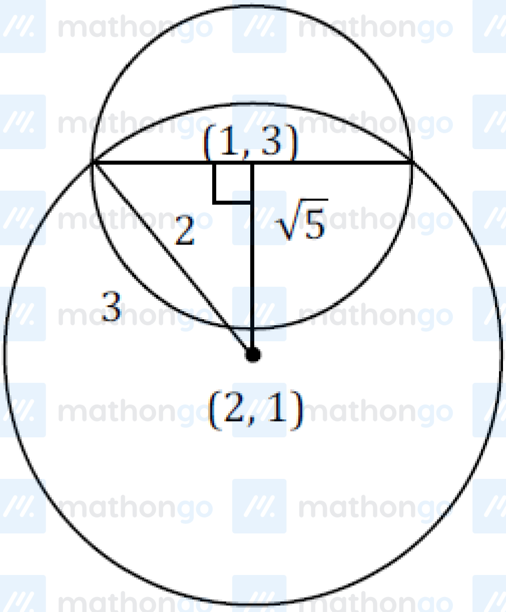
Q1 (3) **Q2** (1225) **Q3** (3) **Q4** (2)

Q1 (3)

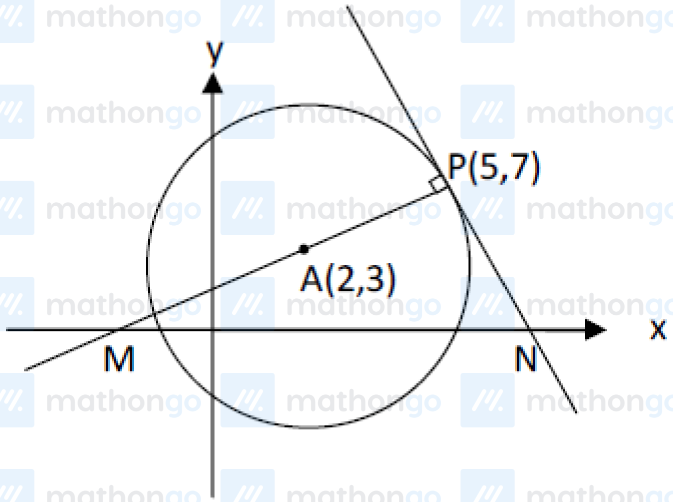
distance between (1,3) and (2,1) is $\sqrt{5}$

$\therefore (\sqrt{5})^2 + (2)^2 = r^2$

$\Rightarrow r = 3$



Q2 (1225)



Equation of normal at P

$(y-7) = \left(\frac{7-3}{5-2}\right)(x-5)$

$3y-21=4x-20$

$\Rightarrow 4x-3y+1=0$ ...(1)

$\Rightarrow \quad M\left(-\frac{1}{4},0\right)$

Equation of tangent at P

$(y-7)=-\frac{3}{4}(x-5)$

$4y-28=-3x+15$

$\Rightarrow 3x+4y=43\ldots\ldots$ (ii)

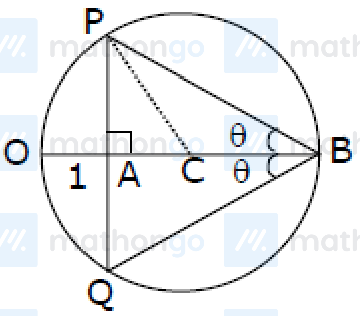
$\Rightarrow \quad \mathrm{N}\left(\frac{43}{3},0\right)$

Hence ar $(\Delta\mathrm{PMN})=\frac{1}{2}\times\mathrm{MN}\times 7$

$\lambda=\frac{1}{2}\times\frac{175}{12}\times 7$

$\Rightarrow 24\lambda=1225$

Q3 (3)



$\mathrm{OC}=\frac{13}{2}=6.5$

$\mathrm{AC}=\mathrm{CO}-\mathrm{AO}$

$=6.5-1$

$=5.5$

In $\Delta\mathrm{PAC}$

$\mathrm{PA}=\sqrt{6.5^2-5.5^2}$

$PA=\sqrt{12}$

$\Rightarrow PQ=2PA=2\sqrt{12}$

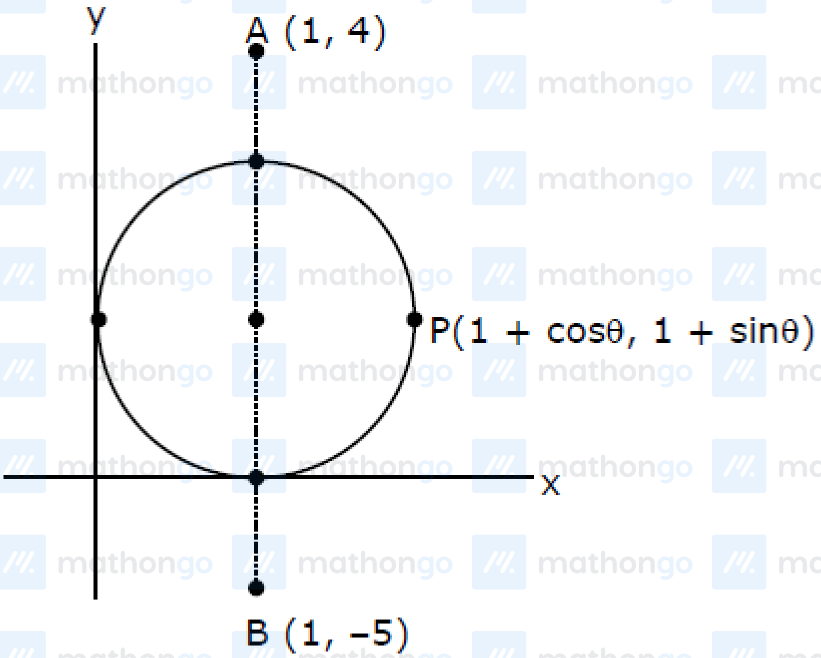
Now, area of $\Delta\mathrm{PQB}=\frac{1}{2}\times\mathrm{PQ}\times\mathrm{AB}$

$= \frac{1}{2} \times 2\sqrt{12} \times 12$

$= 12\sqrt{12}$

$= 24\sqrt{3}$

Q4 (2)



$\therefore \mathrm{PA}^2 = \cos^2\theta + (\sin\theta - 3)^2 = 10 - 6\sin\theta$

$\mathrm{PB}^2 = \cos^2\theta + (\sin\theta - 6)^2 = 37 - 12\sin\theta$

$\mathrm{PA}^2 + \mathrm{PB}^2\big|_{\max.} = 47 - 18\sin\theta\big|_{\min.} \Rightarrow \theta = \frac{3\pi}{2}$

$\therefore$ P, A, B lie on a line $x = 1$